Rishabh Kumar Singh

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Research Interests

Nonlinear control and stability theory for multi-agent systems, especially *indirect control* of non-cooperative agents through repulsive interactions. Provable single-herder strategies for herding, aggregation and containment, built on Lyapunov methods, limit-cycle and bifurcation analysis, and region-of-attraction estimation. Broader interests: pursuit–evasion and differential games, nonholonomic systems, and optimal control.

Education

Indian Institute of Technology Bombay (IIT Bombay) *Mumbai, India*

Ph.D. in Electrical Engineering (Control and Computing) *2021 – present*

- Thesis topic: Herding of multiple non-cooperative agents with a single herder
- Advisor: Prof. Debraj Chakraborty
- Expected thesis submission: December 2026

Indian Institute of Technology (BHU) *Varanasi, India*

M.Tech. in Systems Engineering *2019 – 2021*

- Thesis: Gain Scheduling Based Controller Design for a Synchronous Machine
- Advisor: Prof. Shyam Kamal; Co-advisor: Prof. S. R. Mohanty

MJP Rohilkhand University *Bareilly, India*

B.Tech. in Electrical Engineering *2013 – 2017*

- Final-year project advisor: Prof. Atul Katiyar

Relevant Graduate Coursework

Optimal Control Systems; Nonlinear Dynamical Systems; Multivariable Control Systems; Decentralized Control of Complex Systems; Games and Information; Optimization.

Research Experience

Herding of Multiple Non-Cooperative Agents with a Single Herder *IIT Bombay*

Doctoral Research, Advisor: Prof. Debraj Chakraborty *2021 – present*

- Formulated the single-herder, multiple-agent herding problem under an inverse-square repulsion law, with no attraction between agents and no "chased/unchased" distinction. Prior single-herder solutions relied on that distinction through switched sliding-mode control.
- Proved that an *open-loop* constant-angular-velocity circular herder trajectory herds *any* number of agents into an arbitrarily small stable limit cycle around the target. Derived stability conditions, closed-form limit-cycle radii and convergence-rate trade-offs, and estimated the region of attraction with quadratic Lyapunov functions [C1].
- Designed a *feedback* herder law in which the herder circles the target at constant angular velocity and adjusts its radius using the radial position of a single agent. Classified all closed-loop equilibria as stable or saddle-type and identified the saddle-node bifurcation values. Proved aggregation for every positive angular velocity, and ϵ-containment for every $\epsilon > 0$ under a suitable choice of herder parameters [J1].
- Derived an explicit local region of attraction in the herder-fixed rotating frame, replacing the earlier numerical estimate. For mutually repulsive agents, estimated the size and centroid offset of the herd and proved that no two agents collide in finite time.
- Extended the framework to nonholonomic unicycle herders and agents. Designed a feedback law that stabilizes the herder's circular motion. Gave an exact local stability analysis (Routh–Hurwitz) of the resulting equilibrium orbits, with no large-heading-gain approximation, and a herding guarantee around the target. For mutually repulsive unicycles, proved that no two agents meet in finite time while their heading errors stay below 90°.

Gain Scheduling Based Controller Design for a Synchronous Machine *IIT (BHU) Varanasi*

M.Tech. Thesis, Advisors: Prof. Shyam Kamal, Prof. S. R. Mohanty *2019 – 2021*

- Posed an optimal control problem for a seven-state synchronous generator model with exciter and power system stabilizer. Computed optimal trajectories from Hamiltonian necessary conditions with the steepest-descent algorithm.
- Linearized the nonlinear dynamics about the optimal trajectory to obtain a linear time-varying system. Used an equivalence (Lyapunov) transformation to design a state-feedback gain that renders the closed loop equivalent to an arbitrary time-invariant system.

Publications

Conference Proceedings

[C1] **R. K. Singh** and D. Chakraborty, "Planar herding of multiple evaders by a single pursuer," in *Proc. 63rd IEEE Conference on Decision and Control (CDC)*, Milan, Italy, 2024, pp. 7375–7380, doi: 10.1109/CDC56724.2024.10886844. Presented by R. K. Singh.

Journal Manuscripts Under Review

[J1] **R. K. Singh** and D. Chakraborty, "Planar herding of multiple agents with a single herder," submitted to *IEEE Transactions on Automatic Control*, May 2025; revised and resubmitted, June 2026. Preprint of an earlier version: arXiv:2505.10048.

Manuscripts in Preparation

Two journal manuscripts on multi-robot herding, in preparation.

Awards & Funding

2024 **Student Travel Support**, 63rd IEEE Conference on Decision and Control (CDC), Milan, Italy

2024 **Conference Travel Grant**, IIT Bombay; full funding to present [C1] at CDC 2024

2021 – present **Ph.D. Teaching Assistantship**, Ministry of Education, Government of India

2019 – 2021 **M.Tech. Teaching Assistantship**, Ministry of Education, Government of India

Technical Expertise

Theory: Lyapunov stability and region-of-attraction estimation; limit cycles and bifurcation analysis; linearization and local stability (Jacobian, Routh–Hurwitz); nonholonomic systems; optimal control (Pontryagin necessary conditions, steepest descent); linear time-varying systems and equivalence transformations; pursuit–evasion problems.
Computation and tools: MATLAB (numerical simulation, optimization with `fmincon`), Simulink; Python; ROS 2, Gazebo; LaTeX.
Experimental validation: Multi-robot experiments on wheeled robot platforms with overhead-camera localization.

Teaching Experience

Department of Electrical Engineering, IIT Bombay *Mumbai, India*

Teaching Assistant *2021 – 2026*

- Control Systems Lab (EE 324): conducted vivas and graded lab experiments.
- Lecture courses: Control Systems (EE 302), Matrix Computations (EE 636), Multivariable Control Systems (EE 640), Nonlinear Dynamical Systems (EE 613), Applied Linear Algebra (EE 635): Conducted tutorials and graded assignments and exams.

Professional Service

Reviewer: IEEE Conference on Decision and Control (CDC) 2024; Indian Control Conference (ICC) 2025.